

Assignment 4

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32355

Q1.a) $\Sigma = \{0, 1\}$
 $1[0-1]^*$

c) Accepted: 1, 10111001
Rejected: 0, 01001

Q7.a) $\Sigma = \{x, y\}$
 $x[x-y]^*y$

c) Accepted: xy, xxyyxy
Rejected: x, xyx

Q2.a) $\Sigma = \{a, b\}$
 $[a-b]^*ab$

c) Accepted: ab, baaab
Rejected: a, aba

Q8.a) $\Sigma = \{0, 1\}$
 1^*

c) Accepted: ^{an}empty string, 1111
Rejected: 0, 1001111

Q3.a) $\Sigma = \{0, 1\}$
 $[0-1]^*00[0-1]^*$

c) Accepted: 00, 1010011
Rejected: 1010101, 1111

Q9.a) $\Sigma = \{a, b\}$
 $[a-b][a-b][a-b]^*$

c) Accepted: ba, aaab
Rejected: a, empty string

Q4.a) $\Sigma = \{a, b\}$
 b^*ab^*

c) Accepted: a, bbabbb
Rejected: bb, bababb

Q10.a) $\Sigma = \{0, 1\}$
 $0[0-1]^* + [0-1]^*1$

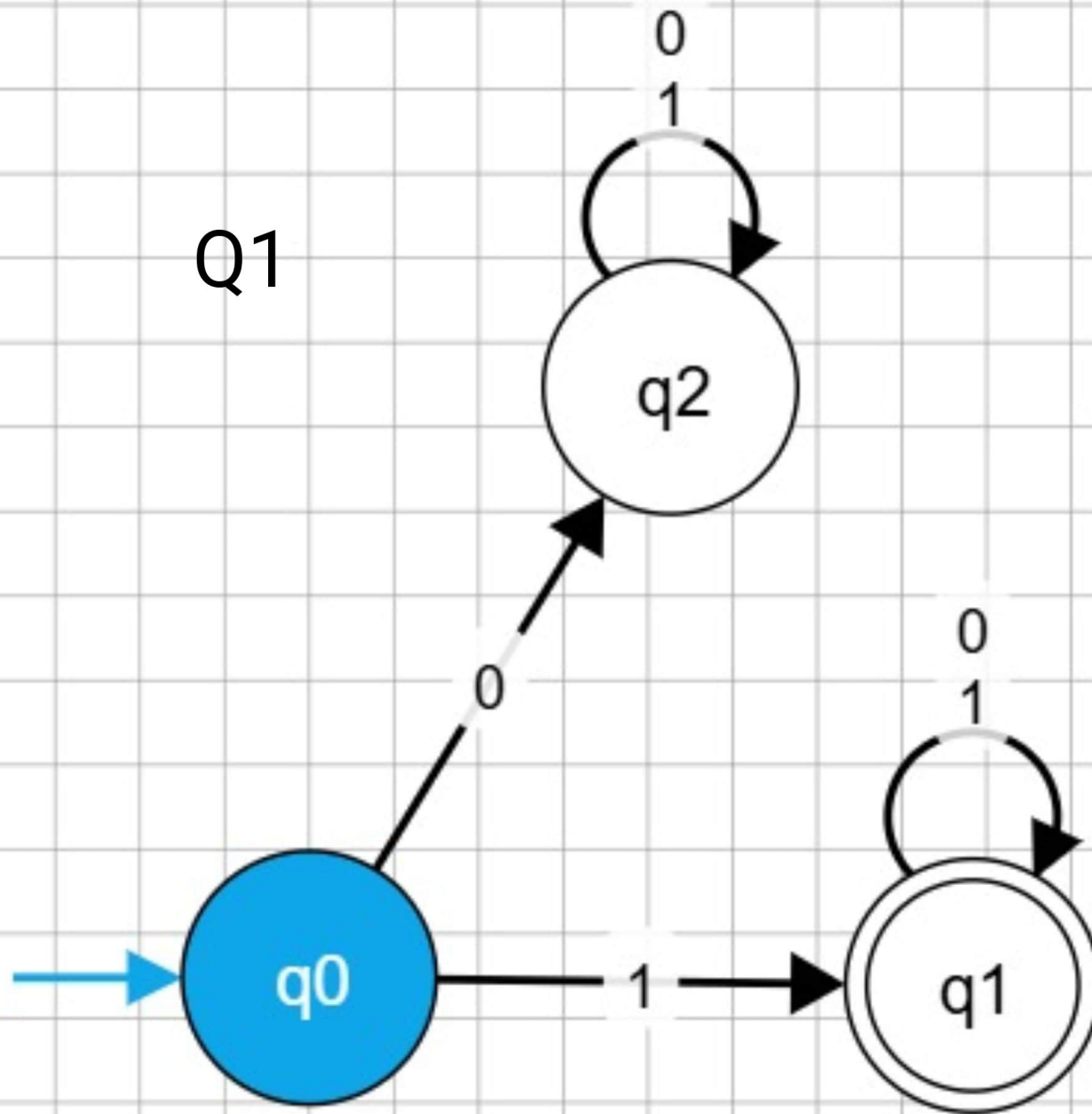
c) Accepted: 0, 0101
Rejected: 10, 111000

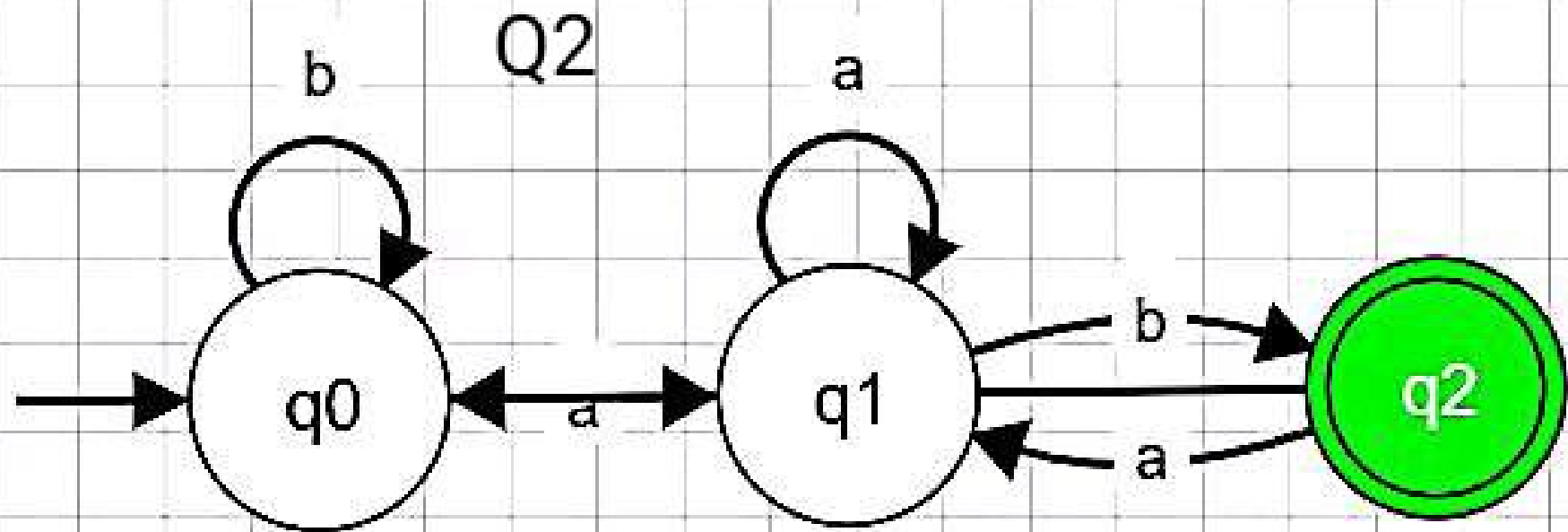
Q5.a) $\Sigma = \{0, 1\}$
 $([0-1][0-1])^*$

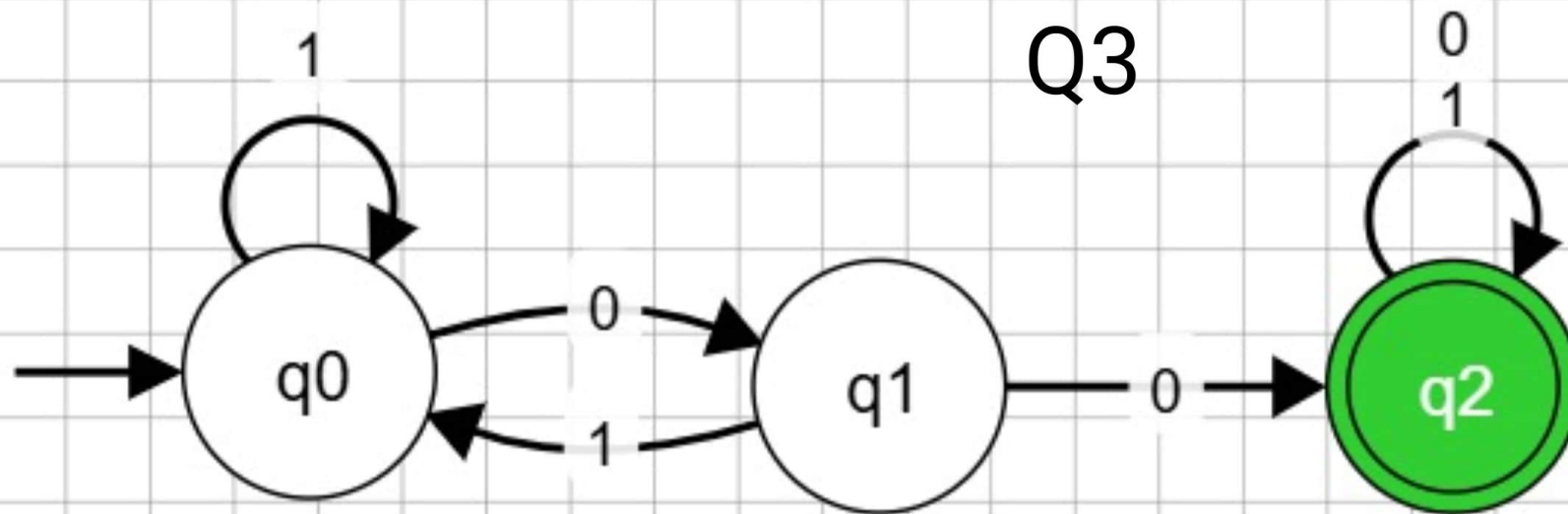
c) Accepted: 10, 1011
Rejected: 0, 101

Q6.a) $\Sigma = \{0, 1\}$
101

c) Accepted: 101
Rejected: 10, 1011

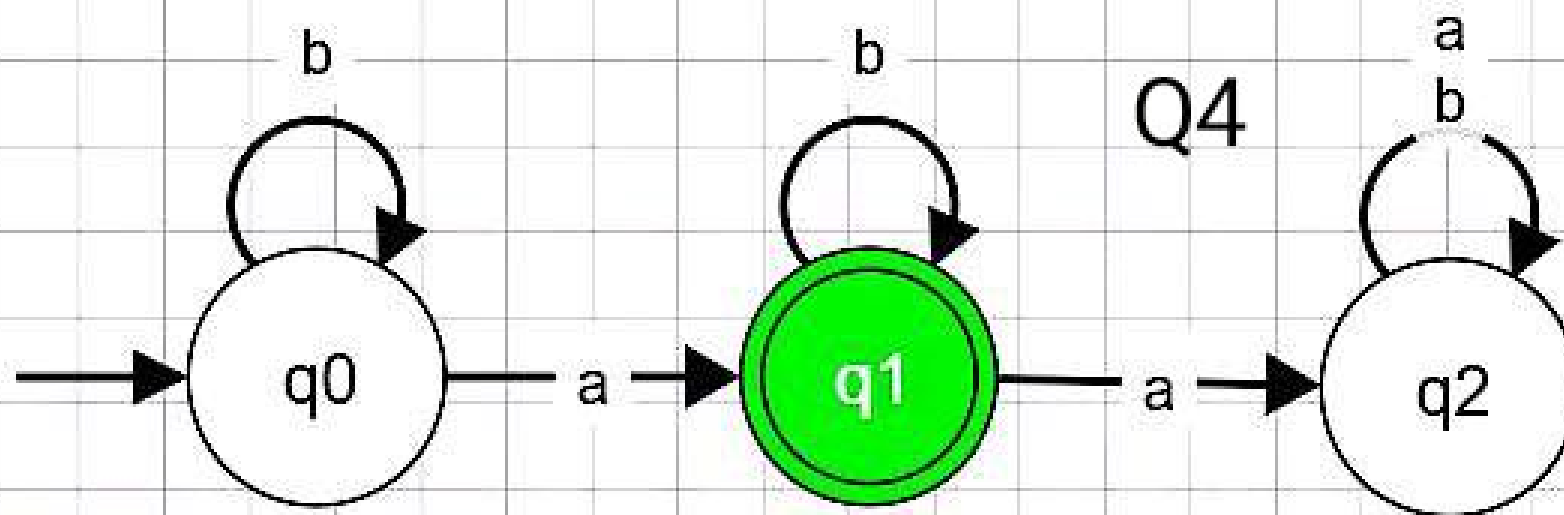




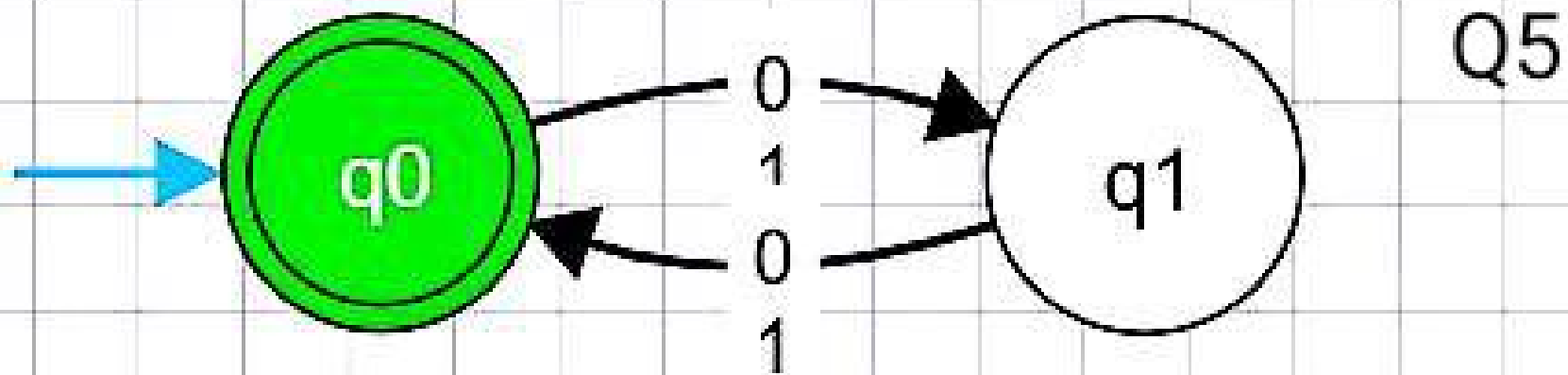


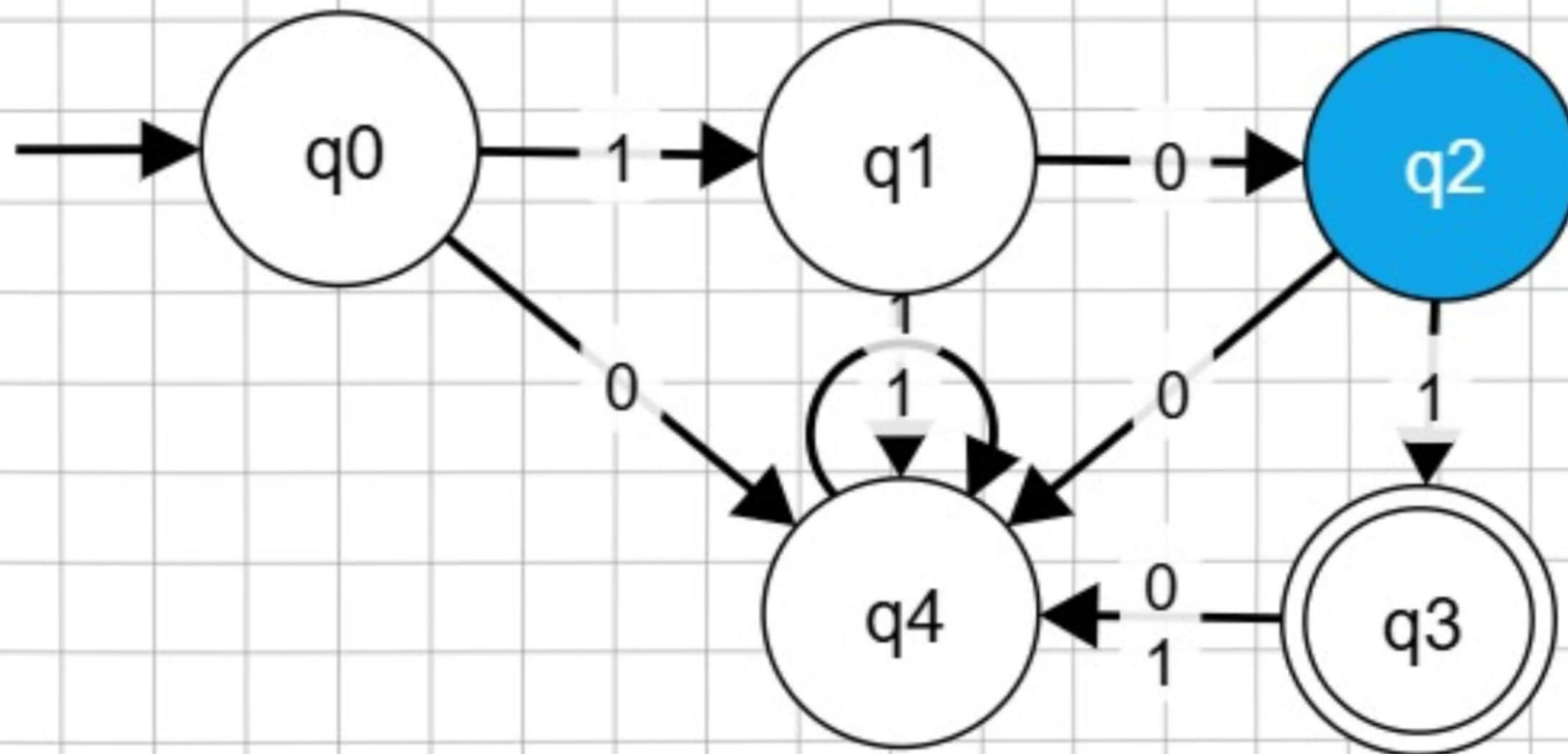
⋮ Tape 1

1	0	1	0	0	1
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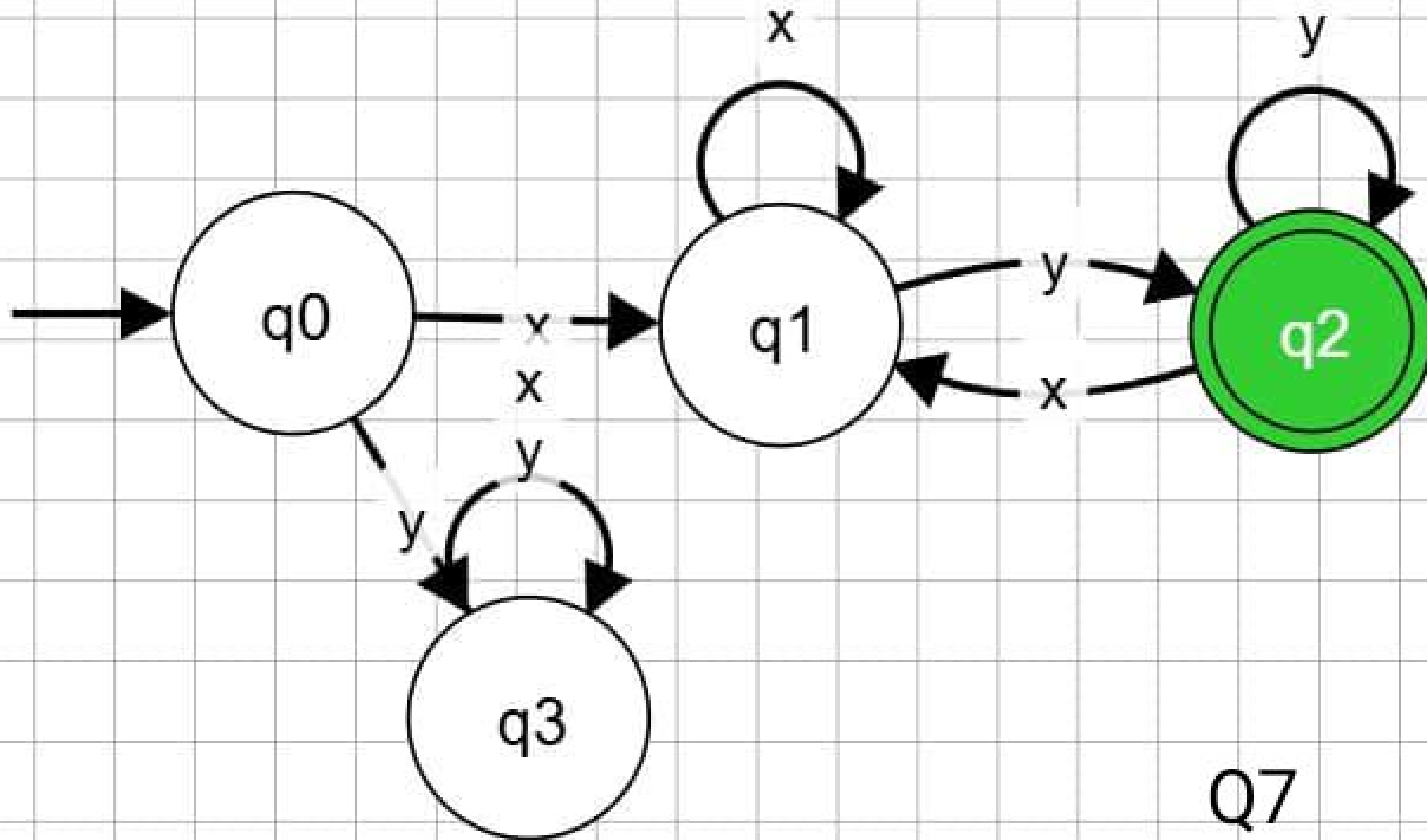


Q4

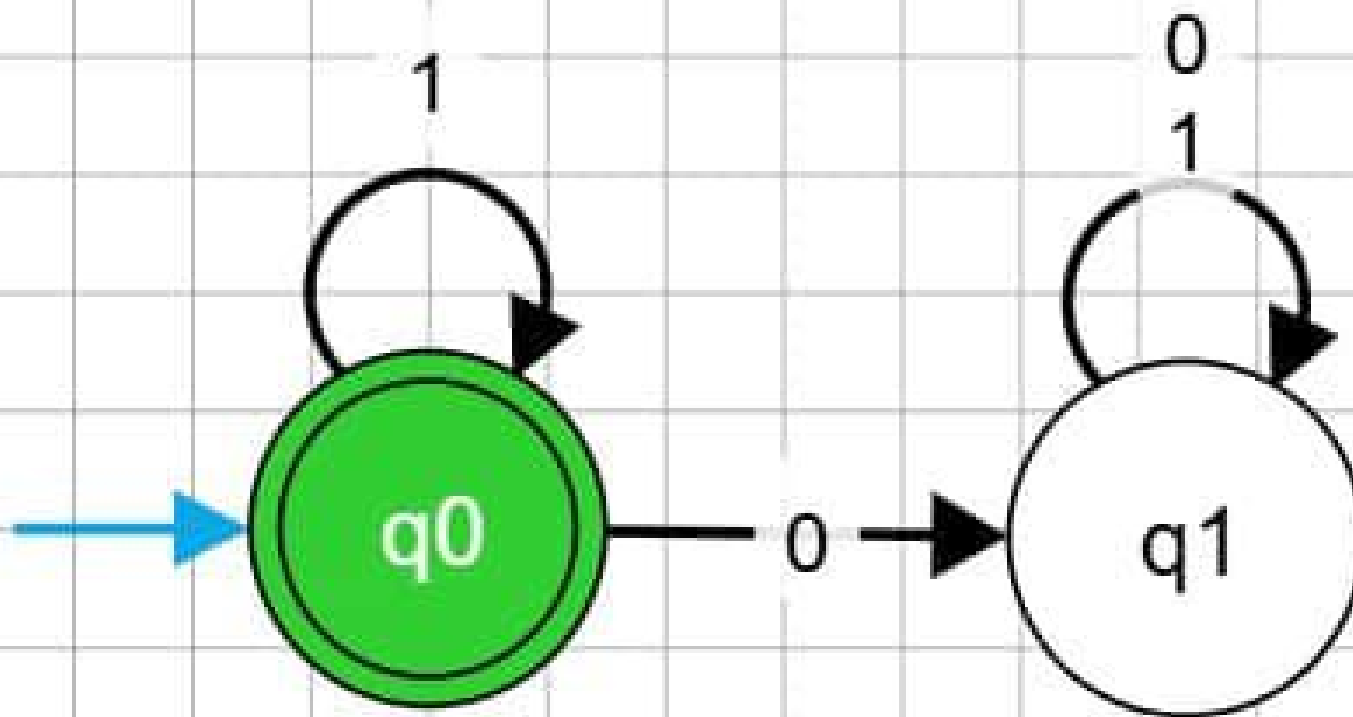




Q6



Q7



Q8

